

Unit 8 - Surface Area and Volume

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The following B.E.S.T Standards will be covered in this unit:

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.

- *Clarification 1: Problem types include finding the volume or a missing dimension of a rectangular prism.*

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's nets.

- *Clarification 1: Instruction focuses on representing a right rectangular prism and right rectangular pyramid with its net and on the connection between the surface area of a figure and its net.*
- *Clarification 2: Within this benchmark, the expectation is to find the surface area when given a net or when given a three-dimensional figure.*
- *Clarification 3: Problems involving right rectangular pyramids are limited to cases where the heights of triangles are given.*
- *Clarification 4: Dimensions are limited to positive rational numbers.*

Unit 8, Lesson 1: Volume of Right Rectangular Prisms



Benchmark

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.



Learning Target

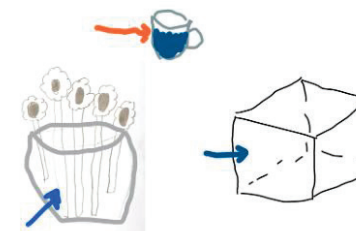
- ☐ I can pack a rectangular prism using a cube with fractional edges.



8.1.1 Warm-Up: Guess the Word

Micah and his family are playing a game in which each person chooses a card with a word written on it. Once a person chooses, they have one minute to draw a picture of the word. Then, his team guesses the word.

Micah chose the word “volume”. His drawing is shown.

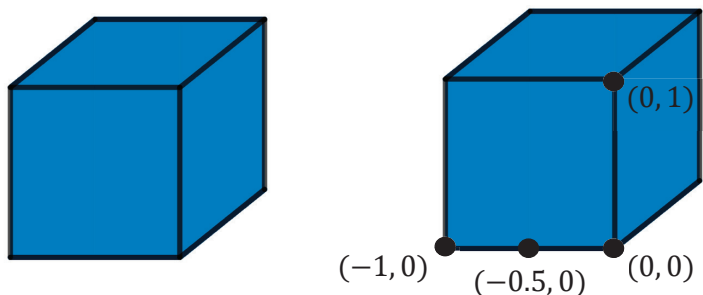


- Choose the statement(s) that best describe volume. Select all that apply.
 - ☐ the sum of the length of all the edges of a three-dimensional object
 - ☐ the distance around a two-dimensional shape
 - ☐ the amount of three-dimensional space enclosed within a container
 - ☐ the total area of the surface of a three-dimensional object
 - ☐ the formula length times width times height



8.1.2 Exploration: The Unit Cube

1. Two images of the same cube are shown. The image on the right shows the coordinates of some of the edges of the cube.



Share three things you know, two things you are not sure of, and one thing you have a question about.

I Know...	I'm Not Sure Of...	A Question I Have...

2. This cube is known as a unit cube because each dimension (length, width, and height) is 1 unit.

What is the volume of the cube? Include the units for the cube's volume.



A **cube** is a rectangular prism with six congruent square faces.



8.1.3 Exploration: Drawing Fractional Cubes

1. Two cubes are shown.

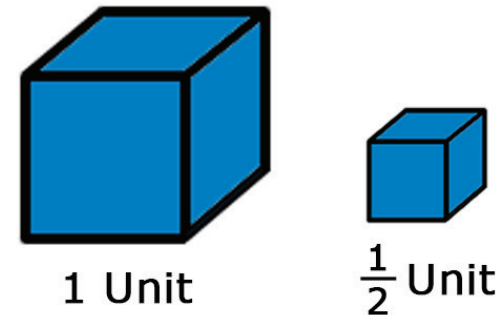


The cube on the left is a unit cube. The cube on the right has different dimensions.

Share an observation and something you wonder about this new cube.

Observation	Wondering

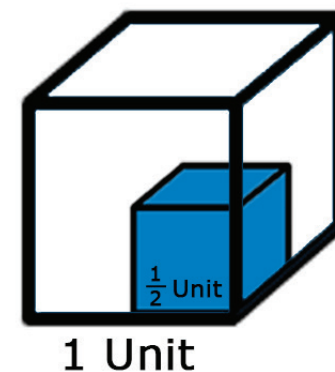
A unit cube and a half-unit cube are shown.



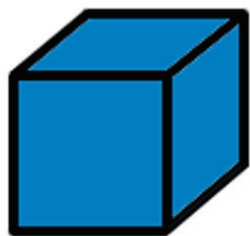
Make a guess of how many cubes with a length of $\frac{1}{2}$ unit, width of $\frac{1}{2}$ unit, and height of $\frac{1}{2}$ unit will fit in the 1-unit cube.

2. The figure shows a half-unit cube inside a unit cube.

Explain how the figure helps you to visualize how to pack cubes that have a fractional length.



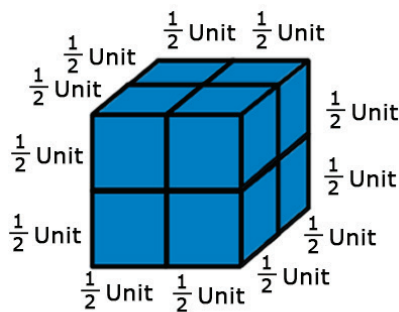
3. Three figures are shown.



1 Unit



$\frac{1}{2}$ Unit



a. Explain how the figures help you visualize how to pack cubes that have a fractional length.

b. To find the number of cubes whose dimensions are $\frac{1}{2}$ unit that fit into a cube whose dimensions are 1 unit, Derek wrote 1 as $\frac{2}{2}$ so he could see that $\frac{1}{2} + \frac{1}{2}$ would add to length of $\frac{2}{2}$ or 1. Complete his explanation.

Since $\frac{1}{2} + \frac{1}{2} = \frac{2}{2}$, or 1, I need _____ cubes for the length, _____ cubes for the width, and _____ cubes for the height.

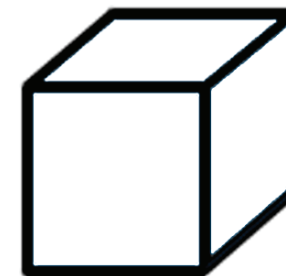
Since _____ $\times$ _____ $\times$ _____ = 8 I know that I need 8 cubes with dimensions $\frac{1}{2}$ unit to fit into one 1-unit cube.



8.1.4 Guided Instruction: Volume with Fractional Cubes

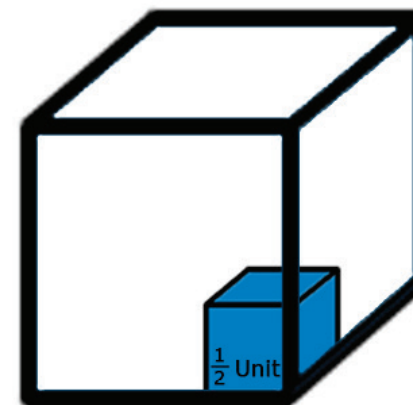
The cube shown is a scale model of a moving box.

1. The edges of the box are $1\frac{1}{2}$ feet long. How many cubes, whose dimensions are $\frac{1}{2}$ feet, can fit inside each dimension of the moving box?



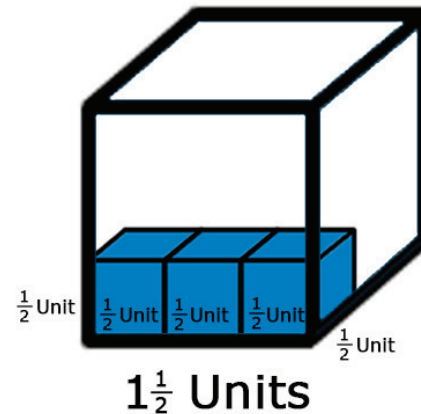
$1\frac{1}{2}$ Units

2. Sketch the cubes that can fit inside each dimension of the moving box.

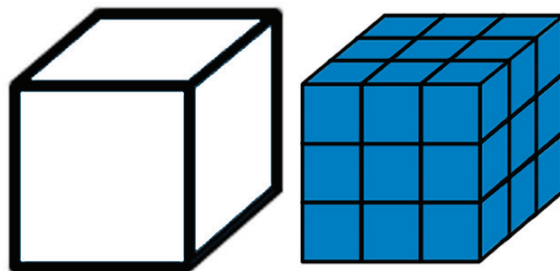


$1\frac{1}{2}$ Units

3. Since 3 cubes, whose dimensions are $\frac{1}{2}$ unit, will fit inside each dimension, how many of these cubes would be needed to pack the moving box?



4. This cube is a scale model of a moving box. The edges of the box are $1\frac{1}{2}$ feet long.



Since there are 27 cubes whose dimensions are $\frac{1}{2}$ feet that will fill the moving box, the volume of the moving box can be found by multiplying the volume of the fractional cube by 27. What is the volume of a cube whose dimensions are $\frac{1}{2}$ feet?

5. Since there are 27 cubes whose volume is $\frac{1}{8}$ cubic feet that will fill in the moving box, the volume of the moving box can be found by multiplying 27 and $\frac{1}{8}$ cubic feet. What is the volume of a cube of the moving box?

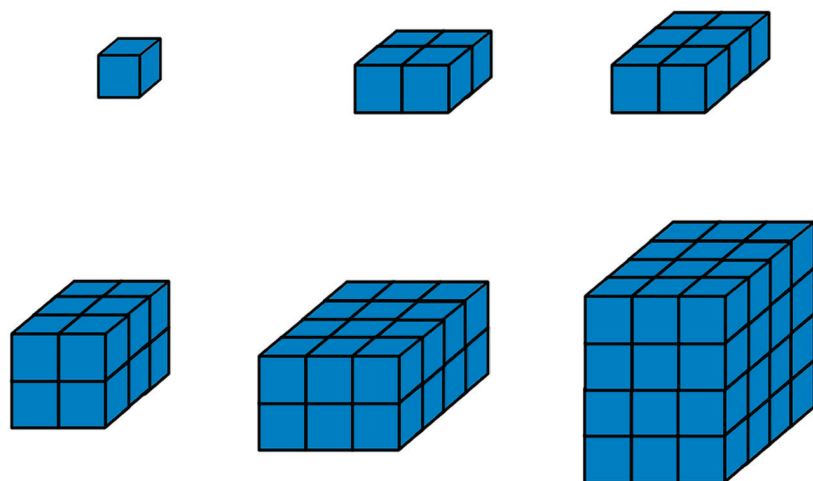
6. Find the volume of the moving box using the formula for volume of a rectangular prism or cube.

$$V = lwh$$

Each dimension of the box is $1\frac{1}{2}$ feet.

7. Does the volume you found using the formula match the volume you found by counting the number of fractional cubes?

8. The images show different rectangular prisms composed of various numbers of cubes.



a. What do you notice?

b. Explain how the dimensions of the prism can be determined using the cubes shown if each small blue cube is 1 unit cube.

9. The diagram shows 3 cubes whose dimensions are $\frac{1}{4}$ inches.



Select the statements that correctly describe the volume of the diagram as shown.

- ☐ Each cube has a volume of $\frac{1}{4}$ cubic inches.
- ☐ Each cube has a volume of $\frac{1}{16}$ cubic inches.
- ☐ Each cube has a volume of $\frac{1}{64}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{4}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{16}$ cubic inches.
- ☐ The volume of all three cubes is $\frac{3}{64}$ cubic inches.

10. The same diagram of 3 cubes whose dimensions are $\frac{1}{4}$ inches is shown.



- a. The dotted line represents the total length of the 3 cubes. How long, in inches, is the dotted line?

- (A) $\frac{4}{4}$ inches
- (B) $\frac{1}{4}$ inches
- (C) $\frac{3}{4}$ inches
- (D) $\frac{2}{4}$ inches

- b. The blue line represents the width of the 3 cubes.

What is the width, in inches?

- (A) $\frac{1}{4}$ inches
- (B) $\frac{2}{4}$ inches
- (C) $\frac{3}{4}$ inches
- (D) $\frac{4}{4}$ inches

- c. The green line represents the height of the cubes.

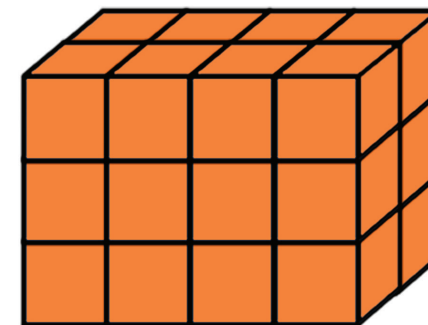
What is the height, in inches?

- (A) $\frac{1}{4}$ inches
- (B) $\frac{2}{4}$ inches
- (C) $\frac{3}{4}$ inches
- (D) $\frac{4}{4}$ inches

- d. Use the dimensions you found to determine the volume of all three cubes with the formula, $V = lwh$.

Finding volume using fractional unit cubes can be done by first finding the volume of one of the fractional unit cubes, and then multiplying that by the total number of fractional unit cubes.

11. A rectangular prism is shown.



- a. How many unit cubes whose dimensions are $\frac{1}{4}$ inch fill up the rectangular prism?

- b. Each unit cube whose dimensions are $\frac{1}{4}$ inch has a volume of $\frac{1}{64}$ cubic inches.

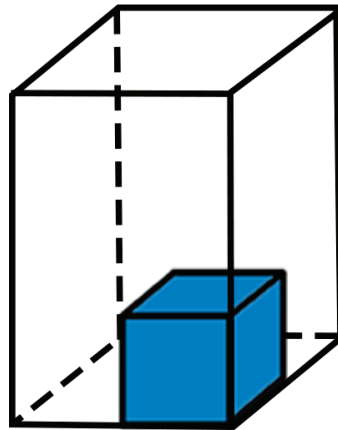
Find the total volume, in cubic inches, represented in the diagram by multiplying the number of cubes that fill the rectangular prism by the volume of one of the cubes.

- c. Find the total volume represented in the diagram, in cubic inches, using the formula, $V = lwh$. Show your work and your answer.



8.1.5 Your Turn!

- A rectangular prism with one cube inside is shown.
 - How many cubes will it take to fill the rectangular prism?



- The rectangular prism has a length of $\frac{16}{3}$ centimeters (cm), width of $\frac{16}{3}$ cm, and height of 8 cm as shown. Use the number of cubes in each dimension to determine the dimensions of the fractional unit cube.

- Complete the statement.

The unit cube shown has dimensions of $\frac{\boxed{}}{3}$ cm. It will take _____ of these unit cubes to fill the rectangular prism.

- Explain how to find the volume of the rectangular prism using the information from Part C.



8.1.6 Wrap-Up: Ticket Out the Door

- Match each image with its corresponding dimensions and volume. Use the table below to record your answers.

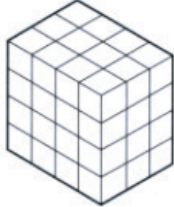
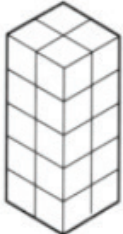
Image	Dimensions	Volume
A  The edges of each cube are $3\frac{1}{4}$ in. long.	D1 Length - 5 in. Width - 5 in. Height - 12.5 in.	V1 Volume $5\frac{1}{3}$ cubic in.
B  The edges of each cube are $\frac{2}{3}$ in. long.	D2 Length - 13 in. Width - 9.75 in. Height - 13 in.	V2 Volume 1,647.75 cubic in.
C  The edges of each cube are 2.5 in. long.	D3 Length - 2 in. Width - $(\frac{4}{3})$ in. Height - 2 in.	V3 Volume 312.5 cubic in.

Image	Dimensions	Volume
A		
B		
C		

Unit 8, Lesson 2: Volume of Rectangular Prisms



Benchmark

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.



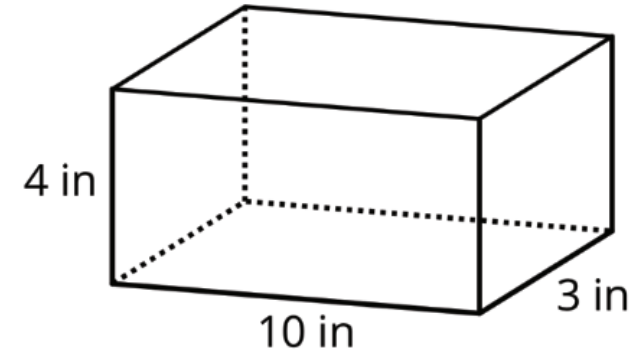
Learning Target

- ☐ I can find the volume of a rectangular prism.



8.2.1 Warm-Up: A Box of Cubes

1. How many cubes with an edge length of 1 inch (in.) fill the box?

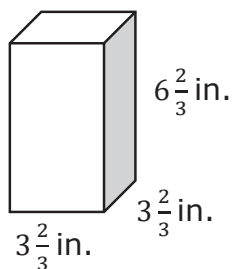


2. If the cubes had an edge length of 2 inches, explain if more or fewer cubes would be needed to fill the box.
3. If the cubes had an edge length of $\frac{1}{2}$ inch, explain if more or fewer cubes would be needed to fill the box.



8.2.2 Guided Instruction: Two Methods for Finding the Volume of a Right Rectangular Prism

Alan calculated the volume of the rectangular prism.



1. Fill in the blanks to complete Alan's statement on how he calculated the volume.

To find the volume of this prism, I will fill the prism with cubes of side lengths $\frac{1}{3}$ in. Because $3\frac{2}{3} \div \frac{1}{3} = \underline{\hspace{2cm}}$, $\underline{\hspace{2cm}}$ cubes will fit along the length and $\underline{\hspace{2cm}}$ cubes will fit along the width. $\underline{\hspace{2cm}}$ cubes will cover the base of the prism. The height of the prism can be used to determine how many layers of cubes are needed. It will take $\underline{\hspace{2cm}}$ layers of cubes because $6\frac{2}{3} \div \frac{1}{3} = \underline{\hspace{2cm}}$. In total, it will take $\underline{\hspace{2cm}}$ of the $\frac{1}{3}$ in. cubes to fill the prism.

The volume of each small cube is $\frac{1}{27}$ in³.

2. Multiply the number of cubes needed to fill the prism by the volume of each cube to find the volume of the prism.

3. Jai used the formula $V = l \times w \times h$ to find the volume of the prism. Complete Jai's work to find the volume of the prism.

$$V = l \times w \times h$$

$$V = \underline{\hspace{2cm}} \times \underline{\hspace{2cm}} \times \underline{\hspace{2cm}}$$

$$V = \underline{\hspace{2cm}} \text{ in}^3$$

4. How are Alan and Jai's methods similar?



The volume, V , of a rectangular prism is

$V = l \times w \times h$, where l represents the length, w represents the width, and h represents the height. This can also be written as $V = B \times h$, where B represents the area of the base and h represents the height.

5. Discuss with your partner how the formula $V = B \cdot h$ is related to the formula $V = l \cdot w \cdot h$.

6. Complete the statements based on your discussion.

The product of the length, l , and the width, w , is equal to _____ of the prism.

$$B = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

Both formulas can be used to find the volume of a right rectangular prism.



8.2.3 Guided Practice: Volume of a Right Rectangular Prism

1. Diego correctly points out that 81 cubes with an edge length of $\frac{1}{3}$ centimeters (cm) are needed to fill a right rectangular prism that has length of 3 cm, width of 1 cm, and height of 1 cm.
 - a. Explain or show how this is true.
 - b. What is the volume, in cubic centimeters, of the rectangular prism?
2. Lin and Noah are packing small cubes into a cube with an edge length of $1\frac{1}{2}$ inches. Lin is using cubes with an edge length of $\frac{1}{2}$ inch, and Noah is using cubes with an edge length of $\frac{1}{4}$ inch.
 - a. Explain who would need more cubes to fill the $1\frac{1}{2}$ in. cube.



- b. Explain whether Lin and Noah will find the same volume of the $1\frac{1}{2}$ inch cube using their method.

- c. What is the volume of the cube with $1\frac{1}{2}$ inch edge lengths? Include units.

3. A toy company is packaging its toys to be shipped. Some small toys are placed inside a cube-shaped box with side lengths of 0.5 inches (in.). These smaller boxes are then packed into a shipping box with dimensions of 12 in. $\times$ 35 in. $\times$ 2.5 in.
- a. How many small toys can be packed into the larger box for shipping?
- b. Use the number of toys that can be shipped in the box to determine the volume of the box. Include units.



8.2.4 Your Turn!

1. To give their animals essential minerals and nutrients, farmers and ranchers often have a block of salt, called a "salt lick" available for their animals. A rancher is ordering a box of cube-shaped salt licks. The edge lengths of each salt lick are $\frac{5}{12}$ foot (ft).



- a. Explain if the volume of one salt lick is greater than or less than 1 cubic foot (ft³).
- b. A box that is $1\frac{1}{4}$ ft $\times$ $1\frac{2}{3}$ ft $\times$ $1\frac{5}{6}$ ft is used to ship the salt licks. How many cubes of salt lick can fit in the box?

2. A pool in the shape of a rectangular prism is being filled with water. The dimensions of the pool are 24 feet (ft) by 15 ft. If the height of the water in the pool is 1.3 ft, what is the volume of the water in cubic feet?



8.2.5 Wrap-Up: Ticket Out the Door

1. Joy keeps her modeling clay in a plastic box that has a length of 21 centimeters (cm), a width of 21 cm, and a height of 63 cm.
 - a. What is the volume of the plastic box? Include units.
 - b. If each piece of cube-shaped clay has a side length of 4.2 cm, how many pieces of clay can be stored in the box?

Unit 8, Lesson 3: Finding the Missing Dimension of a Rectangular Prism



Benchmark

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.



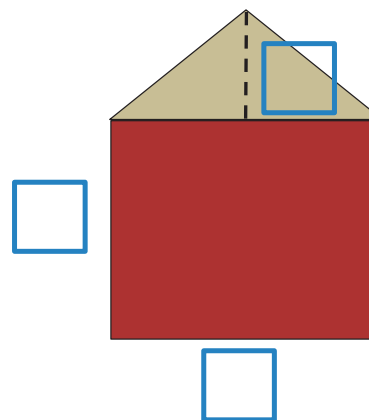
Learning Target

- ☐ I can find the value of a missing dimension of a rectangular prism.



8.3.1 Warm-Up: Area of a Wall

1. Using the digits 0 – 9, at most one time each, fill in the boxes to make a true statement.



Area of  =

Area of  =



8.3.2 Collaboration: Missing Measurement

Jolene works for a textbook company filling orders. She has discovered each textbook has a width of 7 inches (in.), a height of $2\frac{2}{3}$ in., and a length of 10 in.



The textbook company has decided to make the textbooks smaller by reducing the length. The new volume of the textbook will be 168 in^3 .

1. Complete the table.

Old Textbook	New Textbook
$V = l \times w \times h$	$V = l \times w \times h$
$V = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$	168 in^3
$V = \underline{\hspace{1cm}} \text{ in}^3$	$= l \times \underline{\hspace{1cm}} \text{ in.} \times \underline{\hspace{1cm}} \text{ in.}$
	$168 \text{ in}^3 = \underline{\hspace{1cm}} \text{ in.} \times \underline{\hspace{1cm}} \text{ in}^2$

2. Work with your partner to determine the new length of the textbook. Explain your process.



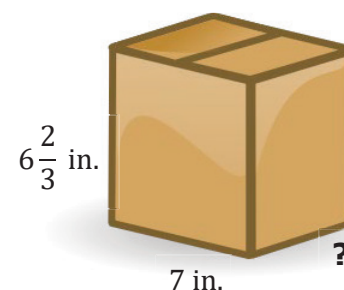
8.3.3 Guided Instruction: Finding a Missing Dimension



Each flat surface of a prism is called a face.

When only two dimensions of a three-dimensional prism are given, the product of the two given dimensions represents the area of that face of the prism.

1. What is the area of the front face of this prism? Include units.



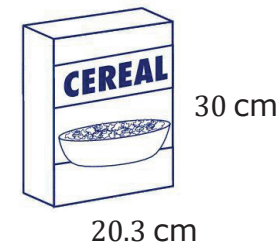
The volume of the prism is equal to the area of the face multiplied by the length of the missing dimension. The value of the missing dimension can then be calculated by dividing the volume by the area of the face.

$$V = \text{Area of the face} \times \text{measurement of the missing dimension}$$

$$\text{Measurement of the missing dimension} = \frac{V}{\text{Area of the face}}$$

2. If the volume of the box is $233\frac{1}{3} \text{ in}^3$, what is the value of the missing dimension of the box?

3. Rafael was trying to find the missing dimension of the cereal box shown.



$$\text{Volume} = 1,218 \text{ cm}^3$$

He made a mistake while solving for the width of the box, but he cannot find it. Inspect his work below.

Rafael's Work

$$\begin{array}{r} 20.3 \text{ cm} \\ \times 30 \text{ cm} \\ \hline 60.9 \text{ cm}^2 \end{array} \quad \text{Area of face}$$

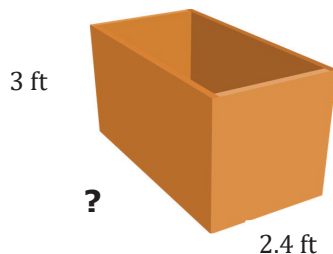
$$\begin{array}{r} 20 \text{ cm}^3 \text{ Volume} \\ 60.9 \overline{) 12180} \\ \underline{1218} \\ 0 \end{array}$$

Write a note to Rafael on how to correct his error(s).



8.3.4 Guided Practice: Find the Missing Dimension

1. Sally broke her ruler before she finished measuring her toy chest. She knows the toy chest has a volume of 25.2 ft^3 .



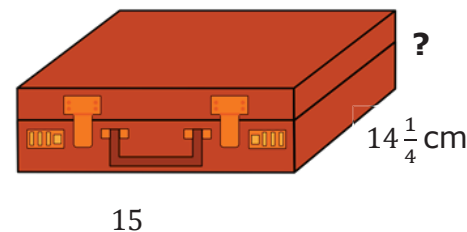
What is the length of the toy chest?

2. Damion works in construction and needs to cut a wooden beam to fill a cement hole. He knows that the beam has a width of 1.2 meters (m) and a length of 0.25 m. If Damion knows that the volume of the beam needs to be 1.5 cubic meters (m^3), what is the height?



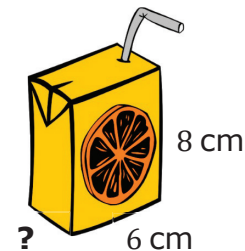
8.3.5 Your Turn!

1. Mr. Rodrigo knows that his briefcase has a volume of $1,068\frac{3}{4}$ cubic centimeters (cm^3). He labeled the dimensions of the briefcase that he knew.



What is the height of the briefcase?

2. Victor works in a factory that makes juice boxes. He knows that the juice boxes have a height of 8 centimeters (cm), a length of 6 cm, and a volume of 192 cubic centimeters (cm^3).



What is the width of the juice box?



8.3.6 Wrap-Up: Reflection

1. Rate your strength on the following areas using the scale.

- 1 – still learning
- 2 – making progress
- 3 – got it

Skill	Rating
Find the volume of a rectangular prism.	
Find a missing dimension of a rectangular prism.	

2. Write an explanation of why you rated yourself the way you did.

Unit 8, Lesson 4: Nets of Rectangular Prisms and Pyramids



Benchmark

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.



Learning Targets

- ☐ I can draw the net of a rectangular prism or rectangular pyramid.
- ☐ I can relate a net to its three-dimensional shape.
- ☐ I can relate a rectangular prism or rectangular pyramid to its net.



8.4.1 Warm-Up: Bell Work

1. A rectangular ball pit has a volume of 28 cubic meters (m^3). Find the width of the ball pit if the length is $5\frac{5}{6}$ m and the depth is $\frac{3}{4}$ m.



8.4.2 Refresher: Area of Composite Figures

1. Work with your partner to find the area of each composite figure. All measurements are in meters (m).

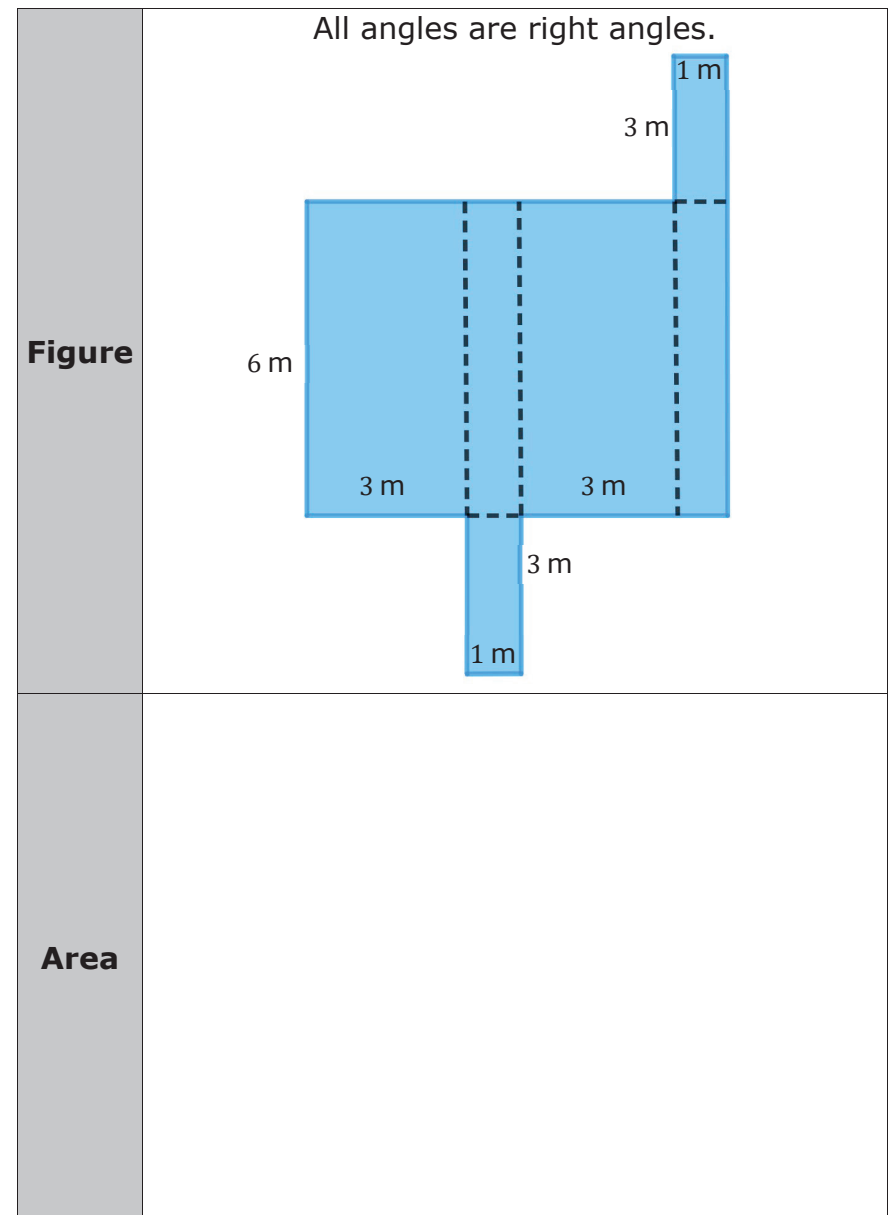
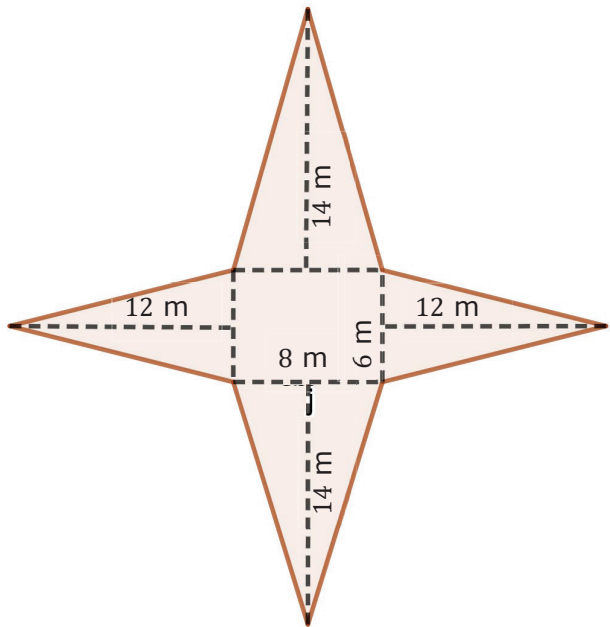


Figure	All dashed lines intersect at right angles. 
Area	



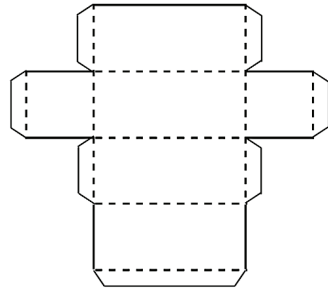
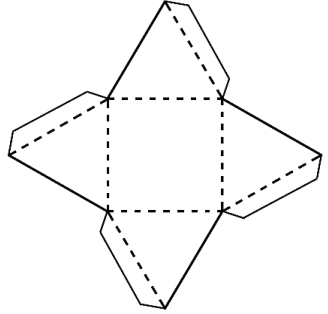
8.4.3 Exploration: Creating Three-Dimensional Figures

Both composite figures in the previous component are nets. A net is formed when the faces of a three-dimensional figure are positioned by arranging the faces to create a two-dimensional composite figure.



A **net** is a two-dimensional diagram that can be folded or made into a three-dimensional figure.

1. Your teacher will provide you with copies of some nets. Cut the nets along the solid line. Fold the nets along the dotted line and use the tabs to glue the figure together.
2. Complete the table to describe the three-dimensional figure formed.

Net	Three-Dimensional Figure Formed
	
	

- Look back at the two composite figures from the refresher. Predict what three-dimensional figure is made from cutting out and folding up the sides of those figures.

- With your partner, discuss how the area of a two-dimensional composite figure is related to the area of the faces of the three-dimensional figure. Summarize your discussion.



8.4.4 Collaboration: Determining Nets

- With your partner, sketch the net or sketch the individual faces of the three-dimensional figure. Label each side of the net with the appropriate measurement in yards (yd).

Three-Dimensional Figure	Net



8.4.5 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on nets?



Unit 8, Lesson 5: Surface Area of Rectangular Prisms and Pyramids



Benchmark

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.



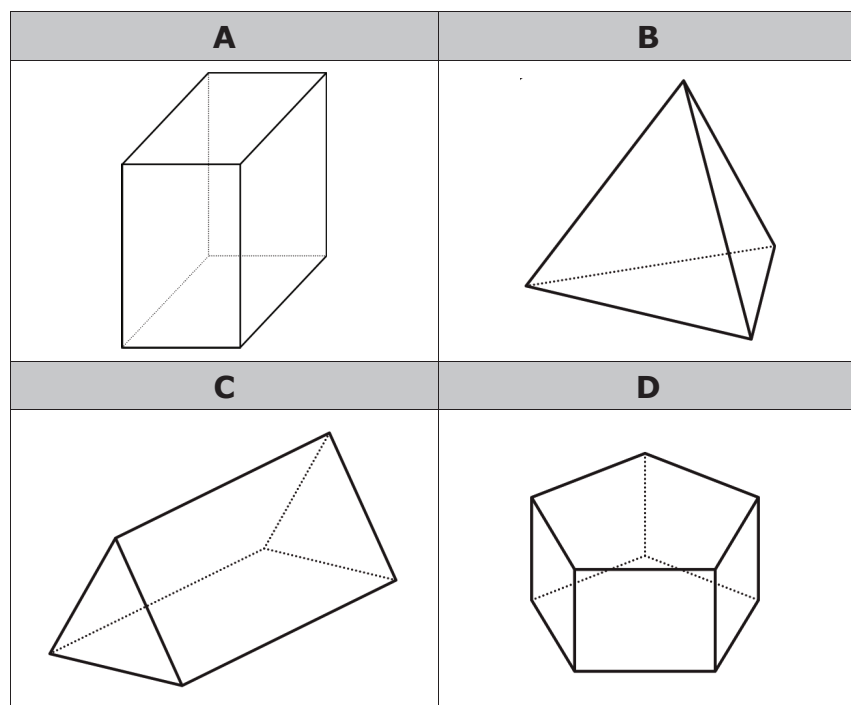
Learning Targets

- ☐ I can find the surface area of a rectangular prism using a net or the figure.
- ☐ I can find the surface area of a rectangular pyramid using a net or the figure.



8.5.1 Warm-Up: Which One Doesn't Belong?

1. Four figures are shown.



- Which one doesn't belong?
- Justify your choice using mathematics.

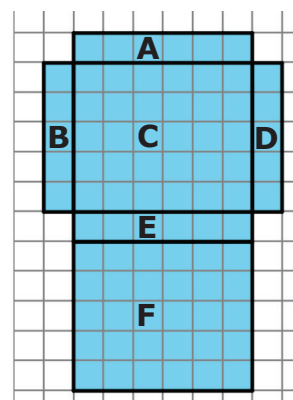


8.5.2 Guided Instruction: Surface Area of a Rectangular Prism

To find the surface area of a three-dimensional object, use the nets of a shape to find the area of each face.

Surface area is the total area of the two-dimensional surfaces that make up a three-dimensional object

- A net of a rectangular prism is shown below with each face labeled.
 - Complete the table to find the area of each face. Each unit of the grid is 1 centimeter (cm).



Face	Dimensions	Area
A		
B		
C		
D		
E		
F		

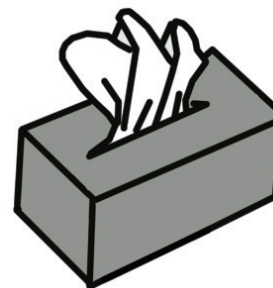
- b. How can the net and the area of the six faces be used to find the surface area of the prism?

- c. What is the surface area of this rectangular prism?



When calculating surface area, find the sum of the areas of each face of the figure. Use the figure's net to label the dimensions needed to calculate the area of each face.

2. A tissue box is 9 inches (in.) long, 5 in. wide, and 6 in. tall.



- a. Sketch the net of the tissue box.

- b. Label the dimensions of each face on the net.

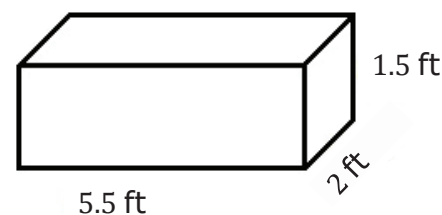
c. Find the area of each face. Include units.

d. Add the area of the faces to find the surface area of the tissue box. Include units.



8.5.3 Collaboration: Analyzing Student Work

- Jillian and Victor are finding the surface area, in square feet (ft^2), of the rectangular prism shown and each got the same answer; however, they solved the problem differently.



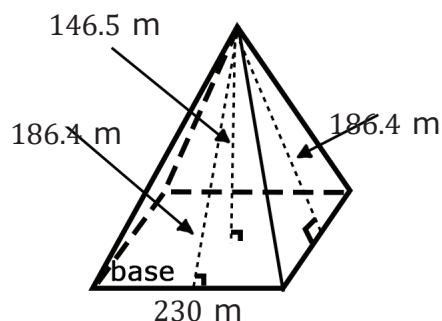
Jillian's Work	Victor's Work
Side 1: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$	2 sides: $2(5.5 \text{ ft} \times 1.5 \text{ ft}) = 16.5 \text{ ft}^2$
Side 2: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$	2 sides: $2(5.5 \text{ ft} \times 2 \text{ ft}) = 22 \text{ ft}^2$
Side 3: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$	2 sides: $2(2 \text{ ft} \times 1.5 \text{ ft}) = 6 \text{ ft}^2$
Side 4: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$	Total Surface Area: 44.5 ft^2
Side 5: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$	
Side 6: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$	
Total Surface Area: 44.5 ft^2	

Discuss with your partner each student's strategy and how they are similar or different. Draw arrows to show connections between each student's work.



8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid

1. Around 2500 B.C. the Egyptians began building pyramids with intricate hallways and layouts to house the tombs of pharaohs and their families. One of the most famous is the Great Pyramid of Giza, also known as the Pyramid of Khufu, seen below. The Egyptians designed the pyramid to have a square base with approximate side lengths of 230 meters, a height of 146.5 meters, and a slant height of 186.4 meters.



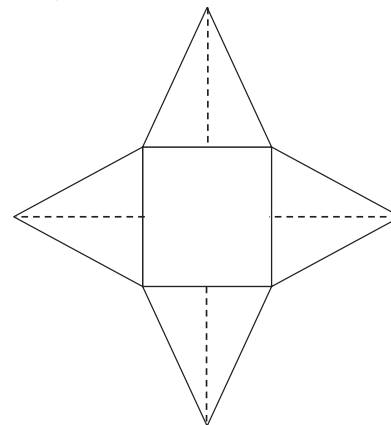
Looking at the model of the pyramid, why is there a measurement called height and a measurement called slant height?

The slant height of the pyramid represents the height of each triangular face.



When finding the surface area of a pyramid, use the **slant height** to find the area of the faces.

2. Using the information about the pyramid, complete the table to determine the surface area of the pyramid. Begin by labeling the sides of the net.



Side	Formula	Area
Rectangular base	$A = l \cdot w$	$A = \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$ $A = \underline{\hspace{2cm}}$
Triangular Face 1	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$ $A = \underline{\hspace{2cm}}$
Triangular Face 2	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$ $A = \underline{\hspace{2cm}}$
Triangular Face 3	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$ $A = \underline{\hspace{2cm}}$
Triangular Face 4	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot \underline{\hspace{2cm}} \cdot \underline{\hspace{2cm}}$ $A = \underline{\hspace{2cm}}$
		Surface Area:

3. Complete the following statement.

To find the surface area of a rectangular pyramid you need

the sum of the ☐ height
☐ area of the base plus the area of

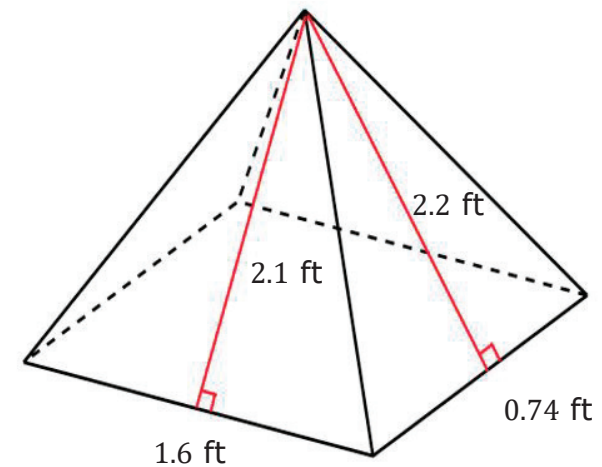
- ☐ one of
☐ all

the other faces.



8.5.5 Collaboration: Surface Area of a Rectangular Pyramid

Papier-mache is a type of art that uses paper and glue. Jose is using the art of papier-mache to create a rectangular pyramid for his art project. To create his project, Jose needs to determine the amount of area that will be covered in newspaper. Below is the drawing Jose created to find the surface area.

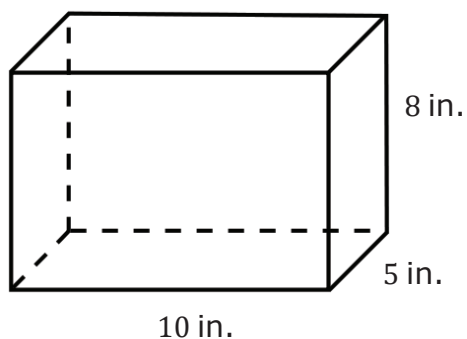


1. Work with your partner to find the surface area of the pyramid.

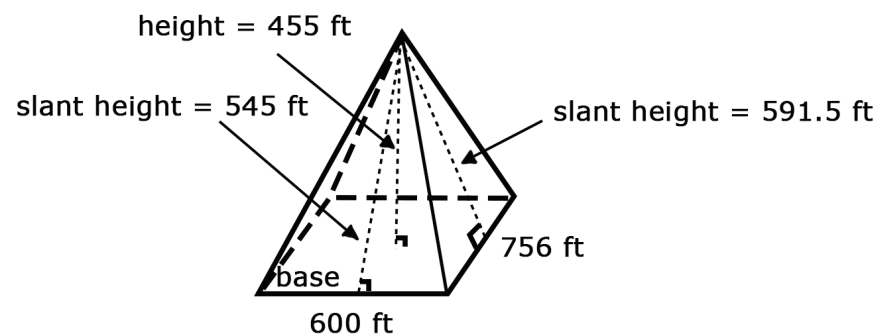


8.5.6 Your Turn!

1. Darsey wants to build trinket boxes made of wood to sell at the local craft festival. A drawing of her box is shown. How much wood does Darsey need to buy for each trinket box?



2. Find the surface area of a rectangular pyramid that has a rectangular base with a side length of 600 feet (ft), which is the base of the triangle with a slant height of 545 ft, and a side length of 756 ft, which is the base of the triangle with a slant height of 591.5 ft.

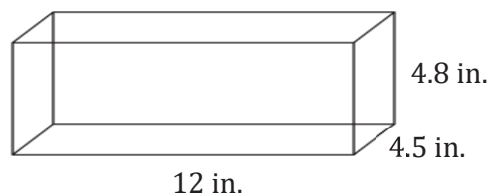




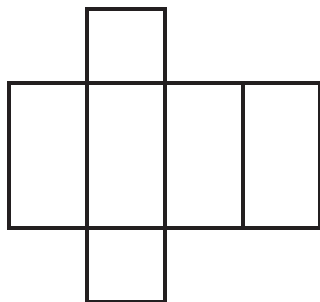
8.5.7 Wrap-Up: Ticket Out the Door

Ted is preparing a holiday display for the post office. He takes an unfolded box and uses it to determine the amount of wrapping paper needed to wrap the box.

The shipping box Ted uses for the display is shown.



1. Write the measurements on the net of the box.



2. Use the net to find the surface area of the shipping box. Include units.



Unit 8, Lesson 6: Finding Surface Area and Volume



Benchmarks

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.

MA.6.GR.2.3 Solve mathematical and real-world problems involving the volume of right rectangular prisms with positive rational number edge lengths using a visual model and a formula.



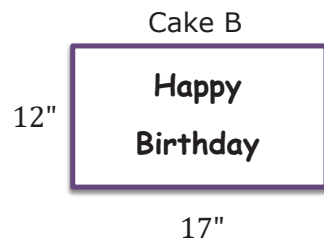
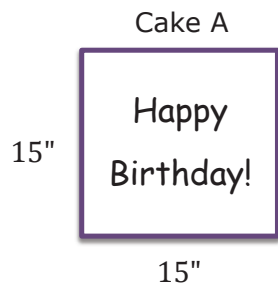
Learning Targets

- ☐ I can find the surface area of right rectangular prisms and right rectangular pyramids.
- ☐ I can find the volume of right rectangular prisms.



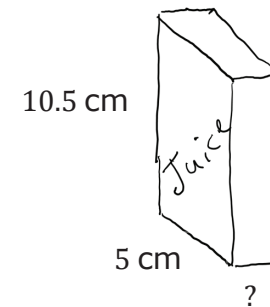
8.6.1 Warm-Up: Would You Rather?

1. Cake A and Cake B are shown below. Cake A is cut into 8 equal slices. Cake B is cut into 6 equal slices. Would you rather have a slice from Cake A or Cake B?



8.6.2 Collaboration: Which Juice Box Is Better?

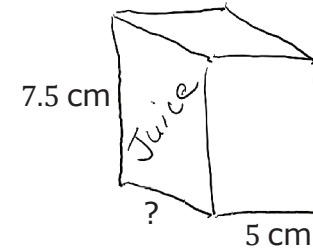
Julius is designing a juice box. He drew a model of his box.



1. If the juice box has a volume of 196.875 cubic centimeters (cm^3), what is the width of the box?

2. How much packaging board would Julius need to make a juice box with these dimensions, not taking into consideration any overlap of the faces?

Julius wondered how the juice box would be affected if he changed the dimensions, but kept the volume the same.



3. What is the new length?
4. How much packaging board would Julius need to make the new juice box?

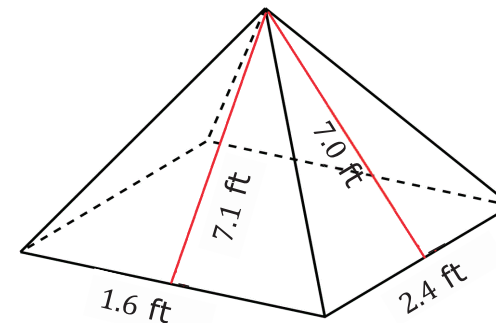
5. If Julius wants to use the least amount of material to construct the juice box, which juice box should he choose? Explain your reasoning.



8.6.3 Guided Practice: Surface Area and Volume

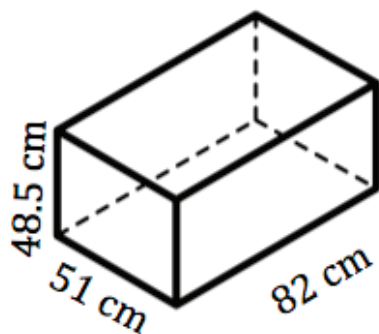
With your partner, complete the chart for the three-dimensional figures below.

1.



What is the type of figure?	
Draw a net or the individual faces of the figure. Label the measurements.	
Find the surface area of the figure.	

2.

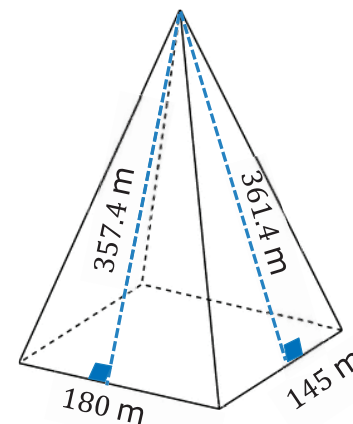


What is the type of figure?	
Draw a net of the figure or the individual faces. Label the measurements.	
Find the surface area of the figure.	
Find the volume of the figure.	



8.6.4 Your Turn!

- Nikolas is constructing a skyscraper with walls made of only glass that is 350 meters (m) high and is in the shape of a rectangular pyramid.

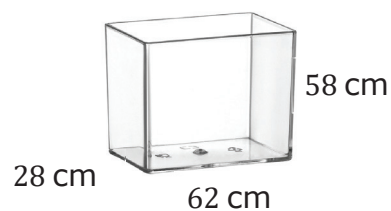


- What is the surface area of the skyscraper?
- Using the model shown, determine how much of the surface area will be made of glass.

2. A cooler has a length of 78 centimeters (cm), a height of 74 cm, and a width of 40 cm.



- Find the surface area of the cooler.
- Find the volume of the cooler.
- If the **inside** of the cooler has a length 62 cm, a height of 58 cm, and a width of 28 cm, how much smaller is the volume of the inside of the cooler than the volume of the entire cooler?



3. Tamyia is creating a gardening box that needs to have a volume of 13.5 ft^3 .

- If the box has a width of 3 feet and a length of 2 feet, what is the height?



- Tamyia decided to paint the exterior walls of the gardening box and needs to know how much paint to buy. What is the surface area of the area she wants to paint?

4. Ryan makes custom jewelry and recently received an order for a pendant in the shape of a rectangular pyramid. Ryan designed a hollow gold pendant to have a square base with a side length of 15 millimeters (mm) and a slant height of 13.24 mm.



How much gold does Ryan need to make the pendant?



8.6.5 Wrap-Up: Reflection

1. Rate yourself on each of the concepts.

- 4 – I can do this on my own.
 3 – I can do this but need more practice.
 2 – I need help getting started.
 1 – I do not understand yet.

Concept	Rating			
I can find the surface area of a right rectangular pyramid using a net or the figure.	1	2	3	4
I can find the surface area of a right rectangular prism using a net or the figure.	1	2	3	4
I can find the volume of a right rectangular prism.	1	2	3	4

2. A barrier to my learning today was ...

3. I will try to overcome this by ...

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